

1. The characteristics of a new aircraft are given in Table B-1. In domestic route service, the applicable regulations require only a 0.75-h fuel reserve for this aircraft. The aircraft has an average route speed of 500 mi/h and an average fuel burn of 13 lb/mi.
 - a. Plot payload versus range for this aircraft.
 - b. Determine the takeoff weight of this aircraft for a 1200-mi trip at maximum payload.
2. An aircraft has the performance characteristics given in Table B-2 in normal route operation. In normal use this aircraft will consume 11,000 lb of fuel in 1 h of flight. The normal cruising speed is 550 mi/h. Generally, alternative destinations are located within 100 mi of planned destinations, and regulations require

TABLE B-1 Preliminary Aircraft Characteristics, lb

Maximum ramp weight	141,000
Maximum structural takeoff weight	140,000
Maximum structural landing weight	128,000
Zero-fuel weight	112,000
Operating empty weight	73,274
Fuel capacity	39,162
Maximum structural payload	38,726
Maximum passenger capacity	27,000
Maximum cargo hold capacity	12,500

TABLE B-2 Aircraft Performance Characteristics, lb

Maximum ramp weight	191,000
Maximum structural takeoff weight	190,000
Maximum structural landing weight	160,000
Zero-fuel weight	140,000
Operating empty weight	100,000
Fuel capacity	55,000
Maximum structural payload	45,000
Maximum passenger capacity	39,000
Maximum cargo hold capacity	15,000

sufficient fuel to fly to the planned destination, fly to an alternate destination, and fly for 1.25 h at the alternate destination.

- a. Find the maximum passenger payload in passengers and lb which may be carried on this aircraft if 15,000 lb of cargo is also carried.
 - b. If a planned route distance is 2000 mi, determine the number of lb of fuel that must be boarded at the gate for this trip.
 - c. If the aircraft carries 180 passengers and no cargo, except for passenger baggage, on a planned 2000-mi trip and if an incident occurs 30 min into the flight, requiring a landing, determine the amount of additional time that the aircraft must remain aloft prior to being able to land safely without exceeding its maximum allowable landing weight.
 - d. Plot payload versus route range for this aircraft.
3. A turbine-powered aircraft has these runway performance characteristics: For a normal takeoff, the liftoff distance is 6500 ft, and the aircraft reaches a height of 35 ft above the end of the runway at a distance of 7000 ft from the runway threshold. For an engine-failure takeoff, the liftoff distance is 7200 ft, and the aircraft reaches a height of 35 ft above the end of the runway at a distance of 7900 ft from the runway threshold. For an aborted takeoff, the accelerate-stop distance is 9700 ft. For landing, the stop distance is 4600 ft. Determine the minimum FAR runway length requirements for this aircraft, indicating the length of the full-strength pavement, the length of the stopway, and the length of the clearway if the runway is to be used by this aircraft in both directions.

5. An airline is considering the use of a particular aircraft on a non-stop route of 1230 mi between two airports. The nearest alternate airport at either destination is 260 mi away. The time fuel reserves are included in the item listed as typical operating empty weight plus time fuel reserves in Table B-3. Table B-4 indicates the takeoff requirements for this aircraft. The airport under study has a runway of 8000 ft available with a maximum difference in centerline elevation of 15 ft. The airport elevation is 500 ft above mean sea level (AMSL), and the mean daily highest temperature in the hottest month is 90°F.
- a. Find the maximum allowable takeoff weight of this aircraft at the study airport.

TABLE B-3 Landing Runway Requirements

Temp., °F	Maximum Allowable Landing Weight, 1000 lb						
	Airport elevation, ft						
	0	1000	2000	3000	4000	5000	6000
80	363.5	363.5	363.5	363.5	363.5	363.5	363.5
85	363.5	363.5	363.5	363.5	363.5	363.5	359.8
90	363.5	363.5	363.5	363.5	363.5	363.5	350.6
95	363.5	363.5	363.5	363.5	363.5	354.6	341.7
100	363.5	363.5	363.5	363.5	358.4	345.5	332.8

Weight, 1000 lb	Landing Runway Length, 1000 ft						
	Airport elevation, ft						
	0	1000	2000	3000	4000	5000	6000
280	5.44	5.57	5.71	5.85	6.00	6.15	6.31
290	5.59	5.73	5.87	6.02	6.17	6.33	6.49
300	5.75	5.89	6.04	6.19	6.35	6.51	6.68
310	5.90	6.05	6.20	6.36	6.52	6.69	6.86
320	6.06	6.21	6.37	6.53	6.70	6.87	7.05
330	6.21	6.37	6.54	6.70	6.87	7.05	7.23
340	6.37	6.53	6.69	6.86	7.03	7.21	7.40
350	6.51	6.66	6.83	7.00	7.19	7.37	7.57
360	6.64	6.81	6.98	7.16	7.35	7.54	7.75
370	6.77	6.96	7.14	7.33	7.52	7.71	7.92

Maximum structural takeoff weight = 408,900 lb

Maximum structural landing weight = 363,500 lb

Typical operating weight plus time fuel reserves = 263,300 lb

Average fuel consumption = 30 lb/mi

Typical maximum passenger load = 54,000 lb

Maximum structural payload = 99,660 lb

Fuel capacity = 21,762 gal

TABLE B-4 Takeoff Runway Requirements

Temp., °F	Airport elevation, ft						
	0	1000	2000	3000	4000	5000	6000
	Maximum Allowable Takeoff Weight, 1000 lb						
80	408.9	400.6	392.9	382.8	369.1	356.2	343.9
85	409.0	400.6	386.0	372.3	359.4	347.1	335.4
90	403.6	388.8	374.9	362.0	349.7	338.1	326.9
95	391.1	377.3	364.2	352.0	340.3	329.2	318.4
100	379.6	366.4	354.1	342.4	331.2	320.4	309.9
	Reference Factor						
80	46.0	48.2	50.5	53.4	57.0	61.0	65.4
85	46.3	48.5	51.8	55.2	59.0	63.1	67.8
90	47.2	50.3	53.6	57.2	61.1	65.5	70.4
95	48.8	52.0	55.4	59.2	63.3	68.0	73.2
100	50.5	53.8	57.4	61.4	65.8	70.7	76.3
	Takeoff Runway Length, 1000 ft						
	Reference factor						
Weight, 1000 lb	40	50	60	70	80		
320	4.23	5.28	6.39	7.50	8.56		
330	4.44	5.58	6.78	7.98	9.09		
340	4.67	5.89	7.18	8.46	9.63		
350	4.90	6.21	7.60	8.95	10.16		
360	5.13	6.55	8.03	9.45	10.68		
370	5.38	6.89	8.46	9.94			
380	5.63	7.25	8.91	10.44			
390	5.89	7.63	9.36	10.92			
400	6.17	8.01	9.82				

- b. If the aircraft takes off from the other airport with a takeoff weight of 338,000 lb with the maximum allowable payload and minimum fuel, find the maximum allowable payload at the origin airport as well as the landing weight and the landing runway requirements at the destination airport.